

Task 2: Data Collection and Preprocessing Report

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Task ID: Task 2 - Data Pipeline

Primary Dataset: California Housing Dataset

Target Domain: Predictive Modeling & Analytics

1. Executive Summary

Data preparation is a foundational pillar in machine learning pipelines. This report documents the rigorous data collection and preprocessing workflow executed for Task 2. Using a robust real-world dataset comprising over 20,000 samples, the project successfully implements automated data cleaning, missing value imputation, duplicate removal, strict data type conversions, outlier management via the Interquartile Range (IQR) technique, and advanced feature engineering to prepare high-quality structured data for modeling tasks.

2. Dataset Description & Collection Source

The selected data source for this portfolio project is the **California Housing Dataset**, originally derived from the 1990 California census data and made publicly accessible via Scikit-Learn.

Dataset Scale

Contains **20,640 samples** (rows) with 8 distinct numerical features and 1 continuous target vector, comfortably exceeding the minimum 1,000 sample threshold requirement.

Features & Attributes

Includes Median Income, House Age, Average Rooms, Average Bedrooms, Population, Average Occupancy, Latitude, and Longitude.

3. Data Collection Pipeline

Data acquisition was programmatic, utilizing standardized public ML repositories to guarantee reproducibility and immediate verification.

- Source Reliability:** Loaded cleanly as a Pandas DataFrame via standard Python APIs (`fetch_california_housing`).
- Integrity Verification:** Inspected structural metadata, feature dimensions, and null distribution maps prior to transformation stages.

4. Data Cleaning & Preprocessing Methodology

To ensure structural consistency and eliminate noise, the raw dataset underwent a multi-stage cleaning pipeline:

Preprocessing Stage	Technique Applied	Execution & Rationale
Missing Values	Median Imputation	Controlled artificial null injections were handled using robust median statistics to prevent skewness.
Duplicate Records	Row Filtering (<code>`drop_duplicates`</code>)	Scanned entire dataframe to remove redundant entries, ensuring unbiased model evaluation.
Type Conversion	Explicit Casting (<code>`astype(float)`</code>)	Enforced consistent numerical floating-point data types across target vectors and feature sets.
Outlier Management	IQR Capping	Applied Interquartile Range thresholds ($\$Q1 - 1.5 \times IQR$ to $\$Q3 + 1.5 \times IQR$) to cap extreme values.

5. Feature Engineering

Advanced feature extraction was performed to enhance predictive signal strength and domain interpretability:

- **New Feature Created:** ``RoomsPerHousehold``
- **Mathematical Formulation:** Derived by dividing ``AveRooms`` by ``AveOccup``.
- **Impact:** Provides a normalized metric reflecting housing space distribution per resident, improving correlation against median house values.

6. Deliverables & Documentation Verification

All project deliverables have been structured and archived in the GitHub repository and verified via Google Drive:

- **Python Script:** ``preprocessing.py`` containing modularized preprocessing functions.
- **Documentation:** Comprehensive ``README.md`` and dataset summary logs.

Status: Completed & Verified. All requirements for Task 2 have been fully satisfied and successfully packaged for review.